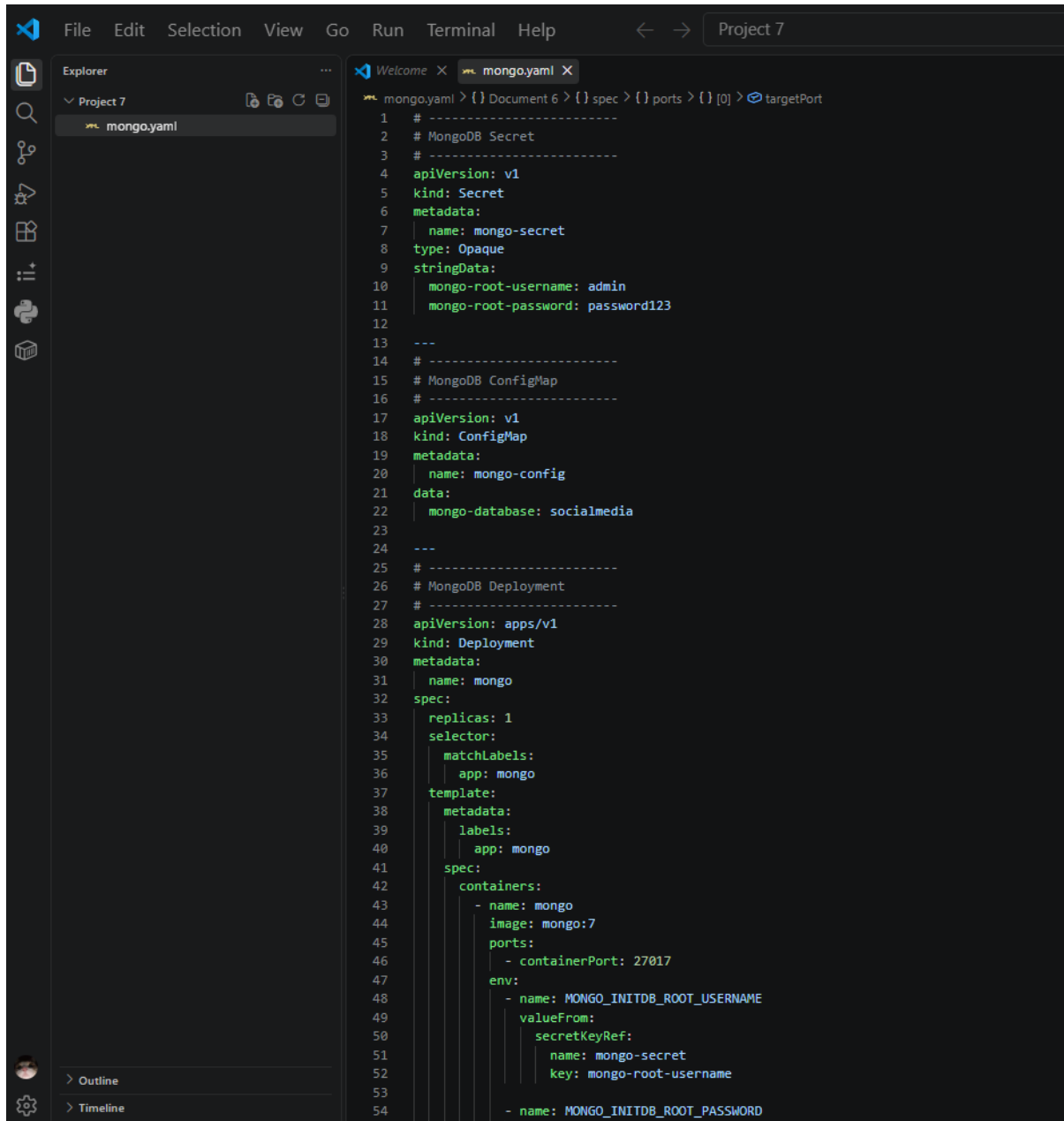


Name: Om Dhamame

PRN: 23070122155



```
1 # -----
2 # MongoDB Secret
3 # -----
4 apiVersion: v1
5 kind: Secret
6 metadata:
7   name: mongo-secret
8 type: Opaque
9 stringData:
10   mongo-root-username: admin
11   mongo-root-password: password123
12
13 ---
14 # -----
15 # MongoDB ConfigMap
16 # -----
17 apiVersion: v1
18 kind: ConfigMap
19 metadata:
20   name: mongo-config
21 data:
22   mongo-database: socialmedia
23
24 ---
25 # -----
26 # MongoDB Deployment
27 # -----
28 apiVersion: apps/v1
29 kind: Deployment
30 metadata:
31   name: mongo
32 spec:
33   replicas: 1
34   selector:
35     matchLabels:
36       app: mongo
37   template:
38     metadata:
39       labels:
40         app: mongo
41     spec:
42       containers:
43         - name: mongo
44           image: mongo:7
45           ports:
46             - containerPort: 27017
47           env:
48             - name: MONGO_INITDB_ROOT_USERNAME
49               valueFrom:
50                 secretKeyRef:
51                   name: mongo-secret
52                   key: mongo-root-username
53             - name: MONGO_INITDB_ROOT_PASSWORD
```

File Edit Selection View Go Run Terminal Help ← → Project 7

Explorer

Project 7

mongo.yaml

Outline

Timeline

Welcome X mongo.yaml X

mongo.yaml > {} Document 6 > {} spec > {} ports > {} [0] > targetPort

28 apiVersion: apps/v1

32 spec:

37 template:

41 spec:

42 containers:

55 valueFrom:

56 secretKeyRef:

57 name: mongo-secret

58 key: mongo-root-password

59

60 ---

61 # -----

62 # MongoDB Service

63 # -----

64 apiVersion: v1

65 kind: Service

66 metadata:

67 name: mongo-service

68 spec:

69 selector:

70 app: mongo

71 ports:

72 - port: 27017

73 targetPort: 27017

74

75 ---

76 # -----

77 # Mongo Express Deployment

78 # -----

79 apiVersion: apps/v1

80 kind: Deployment

81 metadata:

82 name: mongo-express

83 spec:

84 replicas: 1

85 selector:

86 matchLabels:

87 app: mongo-express

88 template:

89 metadata:

90 labels:

91 app: mongo-express

92 spec:

93 containers:

94 - name: mongo-express

95 image: mongo-express:1.0.2

96 ports:

97 - containerPort: 8081

98 env:

99 - name: ME_CONFIG_MONGODB_SERVER

100 value: mongo-service

101

102 - name: ME_CONFIG_MONGODB_PORT

103 value: "27017"

```
79  apiVersion: apps/v1
80  spec:
81    template:
82      spec:
83        containers:
84          - name: ME_CONFIG_MONGODB_PORT
85            value: "27017"
86          - name: ME_CONFIG_MONGODB_ADMINUSERNAME
87            valueFrom:
88              secretKeyRef:
89                name: mongo-secret
90                key: mongo-root-username
91          - name: ME_CONFIG_MONGODB_ADMINPASSWORD
92            valueFrom:
93              secretKeyRef:
94                name: mongo-secret
95                key: mongo-root-password
96
97  ---
98  # -----
99  # Mongo Express Service
100 # -----
101 apiVersion: v1
102 kind: Service
103 metadata:
104   name: mongo-express-service
105 spec:
106   type: NodePort
107   selector:
108     app: mongo-express
109   ports:
110     - port: 8081
111       targetPort: 8081
```

```

PS Z:\0m\SIT\Semester 7\DevOps Lab\Project 7> kubectl apply -f mongo.yaml
secret/mongo-secret created
configmap/mongo-config created
deployment.apps/mongo created
service/mongo-service created
deployment.apps/mongo-express created
service/mongo-express-service created
PS Z:\0m\SIT\Semester 7\DevOps Lab\Project 7> kubectl get all

```

NAME	READY	STATUS	RESTARTS	AGE
pod/backend-54c679dfd9-g6hqb	1/1	Running	1 (33m ago)	24d
pod/flask-deployment-84964d4d6b-2p662	1/1	Running	2 (33m ago)	35d
pod/flask-deployment-84964d4d6b-lbncm	1/1	Running	2 (33m ago)	35d
pod/frontend-cbb6d74c4-cz2bx	1/1	Running	1 (33m ago)	24d
pod/load-generator	1/1	Running	0	4m1s
pod/mongo-58579ff459-mtp4f	0/1	ContainerCreating	0	7s
pod/mongo-express-5c6786d87f-t7t57	0/1	ContainerCreating	0	6s
pod/redis-696c967b87-d7qx2	1/1	Running	1 (33m ago)	24d
pod/social-media-app-5475bb6f57-txnkc	1/1	Running	0	2m27s
pod/social-media-app-5475bb6f57-znnd8	1/1	Running	0	28m

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
service/backend-service	ClusterIP	10.99.250.180	<none>	5000/TCP	24d
service/flask-service	NodePort	10.109.217.59	<none>	5000:30007/TCP	35d
service/frontend-service	NodePort	10.103.10.173	<none>	80:31777/TCP	24d
service/kubernetes	ClusterIP	10.96.0.1	<none>	443/TCP	35d
service/mongo-express-service	NodePort	10.108.217.44	<none>	8081:31340/TCP	7s
service/mongo-service	ClusterIP	10.110.253.176	<none>	27017/TCP	7s
service/redis	ClusterIP	10.99.67.140	<none>	6379/TCP	24d
service/social-media-service	NodePort	10.101.35.224	<none>	80:30405/TCP	28m

NAME	READY	UP-TO-DATE	AVAILABLE	AGE
deployment.apps/backend	1/1	1	1	24d
deployment.apps/flask-deployment	2/2	2	2	35d
deployment.apps/frontend	1/1	1	1	24d
deployment.apps/mongo	0/1	1	0	7s
deployment.apps/mongo-express	0/1	1	0	7s
deployment.apps/redis	1/1	1	1	24d
deployment.apps/social-media-app	2/2	2	2	28m

NAME	DESIRED	CURRENT	READY	AGE
replicaset.apps/backend-54c679dfd9	1	1	1	24d
replicaset.apps/backend-66dd9756bc	0	0	0	24d
replicaset.apps/flask-deployment-84964d4d6b	2	2	2	35d
replicaset.apps/frontend-68756679cb	0	0	0	24d

```
PS Z:\0m\SIT\Semester 7\DevOps Lab\Project 7> kubectl get all
NAME                                     READY   STATUS    RESTARTS   AGE
pod/backend-54c679dfd9-g6hqb            1/1     Running   1 (34m ago) 24d
pod/flask-deployment-84964d4d6b-2p662   1/1     Running   2 (34m ago) 35d
pod/flask-deployment-84964d4d6b-lbncm    1/1     Running   2 (34m ago) 35d
pod/frontend-cbb6d74c4-cz2bx            1/1     Running   1 (34m ago) 24d
pod/load-generator                       1/1     Running   0           5m35s
pod/mongo-58579ff459-mtp4f              0/1     ContainerCreating 0           101s
pod/mongo-express-5c6786d87f-t7t57       0/1     ContainerCreating 0           100s
pod/redis-696c967b87-d7qx2              1/1     Running   1 (34m ago) 24d
pod/social-media-app-5475bb6f57-txnkc     1/1     Running   0           4m1s
pod/social-media-app-5475bb6f57-znnd8    1/1     Running   0           30m

NAME                                     TYPE          CLUSTER-IP   EXTERNAL-IP   PORT(S)          AGE
service/backend-service                 ClusterIP      10.99.250.180 <none>         5000/TCP         24d
service/flask-service                   NodePort       10.109.217.59 <none>         5000:30007/TCP   35d
service/frontend-service                 NodePort       10.103.10.173 <none>         80:31777/TCP     24d
service/kubernetes                       ClusterIP      10.96.0.1     <none>         443/TCP          35d
service/mongo-express-service            NodePort       10.108.217.44 <none>         8081:31340/TCP   101s
service/mongo-service                    ClusterIP      10.110.253.176 <none>         27017/TCP         101s
service/redis                            ClusterIP      10.99.67.140  <none>         6379/TCP         24d
service/social-media-service              NodePort       10.101.35.224 <none>         80:30405/TCP     30m

NAME                                     READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/backend                 1/1     1             1           24d
deployment.apps/flask-deployment        2/2     2             2           35d
deployment.apps/frontend                 1/1     1             1           24d
deployment.apps/mongo                   0/1     1             0           101s
deployment.apps/mongo-express            0/1     1             0           101s
deployment.apps/redis                    1/1     1             1           24d
deployment.apps/social-media-app         2/2     2             2           30m

NAME                                     DESIRED   CURRENT   READY   AGE
replicaset.apps/backend-54c679dfd9       1         1         1       24d
replicaset.apps/backend-66dd9756bc       0         0         0       24d
replicaset.apps/flask-deployment-84964d4d6b 2         2         2       35d
replicaset.apps/frontend-68756679cb      0         0         0       24d
replicaset.apps/frontend-cbb6d74c4       1         1         1       24d
replicaset.apps/mongo-58579ff459         1         1         0       101s
replicaset.apps/mongo-express-5c6786d87f 1         1         0       101s
replicaset.apps/redis-696c967b87         1         1         1       24d
replicaset.apps/social-media-app-5475bb6f57 2         2         2       30m

NAME                                     REFERENCE          TARGETS          MINPODS   MAXPODS   REPLICAS   AGE
horizontalpodautoscaler.autoscaling/social-media-app  Deployment/social-media-app  cpu: 41%/50%    1         5         2         27m
PS Z:\0m\SIT\Semester 7\DevOps Lab\Project 7> kubectl get configmap
PS Z:\0m\SIT\Semester 7\DevOps Lab\Project 7> kubectl get secrets
NAME      DATA   AGE
kube-root-ca.crt  1       35d
```

PS Z:\0m\SIT\Semester 7\DevOps Lab\Project 7> minikube service mongo-express-service

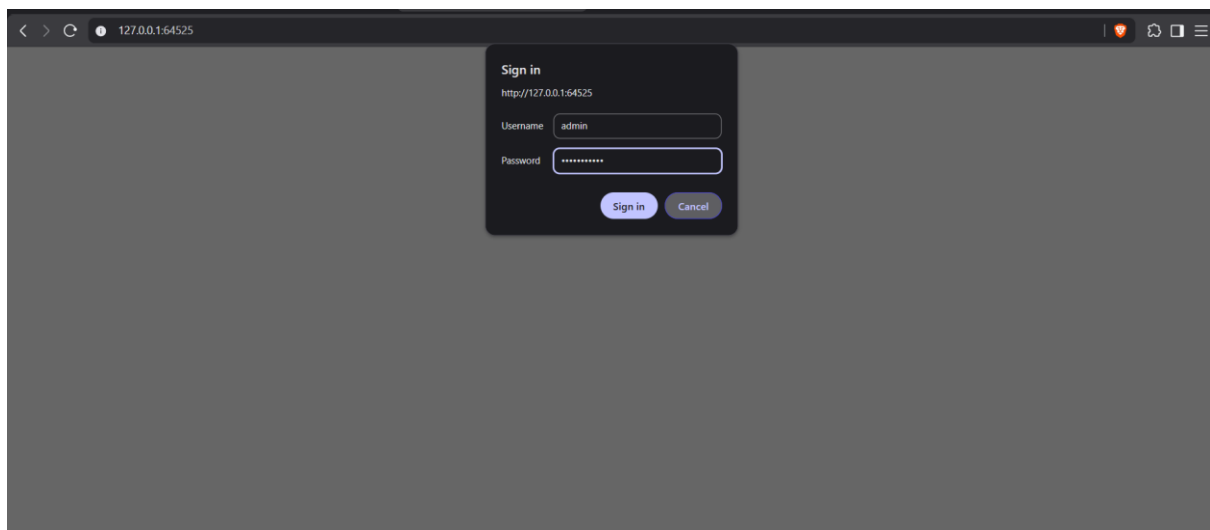
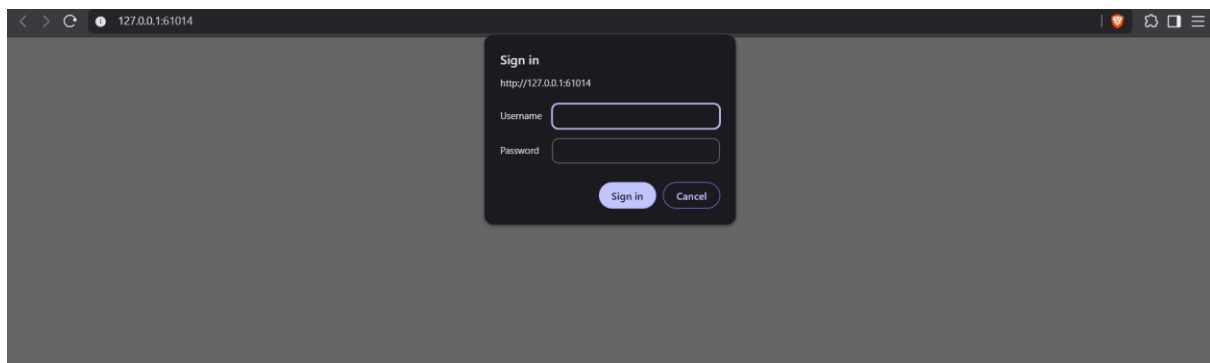
NAMESPACE	NAME	TARGET PORT	URL
default	mongo-express-service	8081	http://192.168.49.2:31340

🔗 Starting tunnel for service mongo-express-service.

NAMESPACE	NAME	TARGET PORT	URL
default	mongo-express-service		http://127.0.0.1:61014

🌐 Opening service default/mongo-express-service in default browser...

! Because you are using a Docker driver on windows, the terminal needs to be open to run it.



Mongo Express Database ▾

Mongo Express

Databases

[+ Create Database](#)

View	admin	Del
View	config	Del
View	local	Del

Server Status

Hostname	mongo-58579ff459-mtp4f	MongoDB Version	7.0.40
Uptime	395 seconds	Node Version	18.20.3
Server Time	Wed, 26 Aug 2026 09:17:20 GMT	V8 Version	10.2.154.26-node.37
Current Connections	3	Available Connections	99683
Active Clients	0	Queued Operations	0
Clients Reading	0	Clients Writing	0

